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A multi-object detection and tracking algorithm for autonomous driving based on improved YOLO26 and OCSORT

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Keywords: autonomous driving, multi-object detection and tracking, YOLO26, OCSORT

Abstract

With the rapid advancement of autonomous driving technology, multi-object detection and tracking has become a critical task in environment perception. To address the challenges of large-scale variation, severe occlusion, strong background interference, and frequent identity switching in complex traffic scenarios, this study proposes a multi-object detection and tracking method based on an improved YOLO26 and OCSORT framework. In the detection stage, the YOLO26 network architecture is optimized by incorporating a more powerful feature extraction module, a multi-scale feature fusion strategy, and an attention mechanism, thereby improving the accuracy, robustness, and real-time performance for detecting traffic participants such as vehicles and pedestrians. In the tracking stage, OCSORT is further enhanced through improvements in motion state estimation and data association, which strengthens the continuity of tracking under occlusion, crossing, and rapid motion conditions, while reducing target loss and identity switches. Experimental results demonstrate that the proposed method achieves improvements in mAP50:95 of 5.60% and 3.40%, and improvements in multiple object tracking accuracy of 3.31% and 2.90% on the KITTI and BDD100K datasets, respectively. Overall, the proposed method outperforms other approaches in terms of accuracy, stability, and real-time performance, and can provide effective technical support for multi-object perception in autonomous driving scenarios.

1. Introduction

With the rapid development of artificial intelligence, computer vision, and intelligent transportation systems, autonomous driving has become an important application scenario of intelligent perception technologies. Autonomous vehicles need to accurately perceive surrounding traffic participants, such as vehicles, pedestrians, cyclists, and other dynamic obstacles, and continuously understand their motion states to support trajectory prediction, path planning, risk assessment, and decision-making control. Therefore, robust object detection and tracking are essential for improving the environmental perception capability, driving safety, and operational stability of autonomous driving systems.

Object tracking is a significant research topic in the field of computer vision, whose primary objective is to continuously estimate and update the spatial position, motion state, and identity information of targets in video sequences [1, 2]. As a fundamental component of video understanding and scene perception, object tracking has been widely applied in video surveillance, intelligent transportation, robotic navigation, unmanned systems, and autonomous driving. In autonomous driving scenarios, object tracking enables vehicles to continuously perceive the motion trajectories and behavioral variations of road participants after detection, thereby providing reliable support for path planning, risk prediction, and decision-making control [3].

Early studies on object tracking were mainly established on state-space models and probabilistic estimation theory, with the central idea of predicting and updating target locations through motion state modeling. Representative methods at this stage include Kalman filtering [4] and particle filtering. Kalman filtering is suitable for target state estimation under linear systems with Gaussian noise, and it offers the

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advantages of high computational efficiency and simple implementation. However, its performance is often constrained in complex scenarios involving nonlinear motion, severe occlusion, and background interference. To address this limitation, particle filtering improves the representation of complex motion patterns and multimodal states by approximating the posterior probability distribution with multiple samples [5], thereby enhancing tracking robustness to a certain extent. Nevertheless, methods in this stage generally rely heavily on prior assumptions about target motion and show limited adaptability to appearance variations and complex scenes.

With the advancement of computer vision, researchers gradually introduced target appearance information into the tracking process, which led to the development of visual tracking methods based on handcrafted features [6]. At this stage, target appearance was typically described using color, texture, edges, and local feature points, while target localization was achieved through techniques such as template matching, mean shift, and correlation filtering [7]. Under specific conditions, these methods demonstrated relatively high real-time performance and promoted the early deployment of object tracking in practical applications [8]. However, owing to the limited representational capacity of handcrafted features, such algorithms often fail to maintain stable performance when targets undergo scale variation, pose change, illumination variation, or partial occlusion, resulting in prominent problems such as identity drift and tracking failure.

The emergence of deep learning has created new opportunities for the development of object tracking. Convolutional neural networks possess powerful feature learning capabilities and can automatically extract discriminative target representations from large-scale data, thereby significantly improving detection and tracking performance in complex scenarios. On this basis, object tracking has gradually evolved toward the mainstream tracking-by-detection paradigm [3], in which candidate targets are first generated by high-performance object detectors and then associated across frames using motion information and appearance features [9]. The SORT algorithm [10] was the first to demonstrate the critical importance of high-quality detection results for multi-object tracking performance, and its high computational efficiency makes it highly practical for real-time applications. However, because SORT relies solely on motion models for data association, it is prone to identity switches in scenarios involving occlusion, target crossing, and target reappearance. To overcome this limitation, DeepSORT [11] further introduced deep appearance features, effectively enhancing identity preservation under occlusion and promoting the development of multi-object tracking toward the integration of detection and re-identification.

In recent years, as the complexity of multi-object tracking tasks has continued to increase, research has gradually shifted toward joint modeling and end-to-end optimization [9]. On the one hand, methods such as FairMOT [12] improve the overall performance of multi-object tracking by unifying object detection and feature extraction within a single framework, thereby alleviating the imbalance between detection and identity recognition tasks. On the other hand, algorithms such as ByteTrack [13] and OC-SORT [14] improve data association mechanisms and state update strategies to address issues such as the loss of low-confidence targets and severe trajectory drift in complex motion scenarios, significantly enhancing tracking stability in dense occlusion, rapid motion, and complex traffic environments. Meanwhile, the introduction of Transformer architectures has endowed models with stronger temporal modeling capability, providing a new research direction for object tracking to evolve from traditional modular processing toward unified modeling and end-to-end optimization [15, 16]. Although end-to-end optimization methods have achieved remarkable progress in multiple tasks and improved the overall integration capability of models, they still face several challenges. Specifically, such methods require a unified model to simultaneously handle object detection and object tracking, which may lead to bottlenecks in both accuracy and real-time performance [17, 18]. In particular, in high-density scenes or fast-motion environments, ensuring the stability and precision of each component remains a challenging technical problem. In addition, the training of end-to-end methods often depends on large-scale datasets and substantial computational resources, while their flexibility for fine-grained task-specific adjustment is relatively limited [19, 20].

The two-stage paradigm decouples object detection and object tracking, allowing each stage to be optimized separately. Such a design ensures that the detection stage emphasizes accurate object localization and classification, whereas the tracking stage focuses on identity maintenance and data association, thereby mitigating the potential conflicts between the two tasks in end-to-end methods [17, 21].

The main contributions of this paper are summarized as follows. First, a lightweight and multi-scale enhanced YOLO26-based detector is developed for autonomous driving scenarios. By improving the feature fusion structure and strengthening small-object representation, the proposed detector improves vehicle detection accuracy under complex traffic scenes while maintaining a compact model scale. Second, an improved OCSORT-based tracking framework is proposed to enhance association robustness

in moving-camera scenarios. By incorporating scale-adaptive motion modeling, camera motion compensation, and track-aware initialization, the tracker improves trajectory continuity and reduces association errors caused by target scale variation and ego-motion. Third, a collaborative two-stage detection-and-tracking framework is constructed by jointly optimizing the detector and tracker. The proposed framework achieves a favorable balance among tracking accuracy, real-time performance, and model complexity, demonstrating strong applicability for multi-object vehicle tracking in autonomous driving scenarios.

2. Method

To address the difficulty of simultaneously achieving high accuracy and real-time performance in multi-object tracking for autonomous driving scenarios, this paper proposes a two-stage framework based on the collaborative optimization of an improved detector and tracker. Built upon a comprehensive analysis of existing methods, the proposed framework achieves high-precision detection and tracking while maintaining a relatively small parameter scale. Current mainstream approaches can generally be divided into end-to-end methods and two-stage 'detection-and-tracking' methods. Although end-to-end methods exhibit advantages in unified modeling, they often suffer from training difficulties, limited generalization capability, and high computational cost in complex scenarios. In contrast, traditional two-stage methods, owing to their decoupled architecture, offer greater flexibility and interpretability; however, their performance is highly dependent on the quality of both the detection and association modules.

Based on this observation, this paper adopts YOLO26 [22] as the object detector and OCSORT as the multi-object tracker, and further improves and optimizes both components to address their respective limitations in vehicle detection and tracking under autonomous driving scenarios.

The proposed multi-object tracking framework comprises five principal components: object detection, trajectory state prediction, three-stage data association, track-aware initialization, and trajectory lifecycle management.

In the detection stage, to overcome the problems of feature detail loss and insufficient detection capability of YOLO26 for small objects, the Neck structure is redesigned by introducing an additional small-object detection head, while DySample [23] is employed to replace the conventional upsampling operation, thereby enhancing multi-scale feature fusion and small-object representation capability. In addition, an LSK-SPPF module is designed to incorporate a large-kernel attention mechanism, which strengthens the global receptive field and spatial attention of the network. Furthermore, ASFF is introduced in the head stage to perform spatially adaptive weighted fusion of four-scale features, thereby improving multi-scale information interaction.

In the tracking stage, an improved Kalman filtering method, termed BSK-KF (Bounding-box Scale-Adaptive Kalman Filter), is proposed. By dynamically adjusting the process noise matrix Q and the observation noise matrix R, the proposed method enhances target motion modeling and trajectory continuity in complex scenarios. Moreover, an ECC [24] (Enhanced Correlation Coefficient)-based camera motion compensation module is introduced to align detection results and trajectory states by estimating the global transformation between consecutive frames, thereby effectively alleviating association errors caused by background motion in moving-camera scenarios. In addition, a TAI [21] (Track-Aware Initialization) module is employed during track initialization. This module treats the positions of matched trajectories as non-removable anchors and combines them with non-maximum suppression (NMS) to constrain and filter the remaining high-confidence detections, retaining only more reliable targets for new track initialization. As a result, the stability and reliability of the tracking process are further improved.

Experimental results demonstrate that the proposed method achieves favorable tracking performance in autonomous driving scenarios, significantly improving overall accuracy while maintaining real-time performance. Compared with methods that rely solely on end-to-end frameworks or unoptimized two-stage approaches, the proposed method exhibits stronger adaptability and greater engineering feasibility in practical applications, thereby providing an effective improvement strategy for multi-object tracking tasks.The improved matching process is shown in figures 1.

3. LDA-YOLO26

3.1. Overall network architecture

Based on the YOLO26 model, this section develops an improved framework, termed LDA-YOLO26, to address the challenges of complex scenes, scale variation, and dense object distributions in autonomous

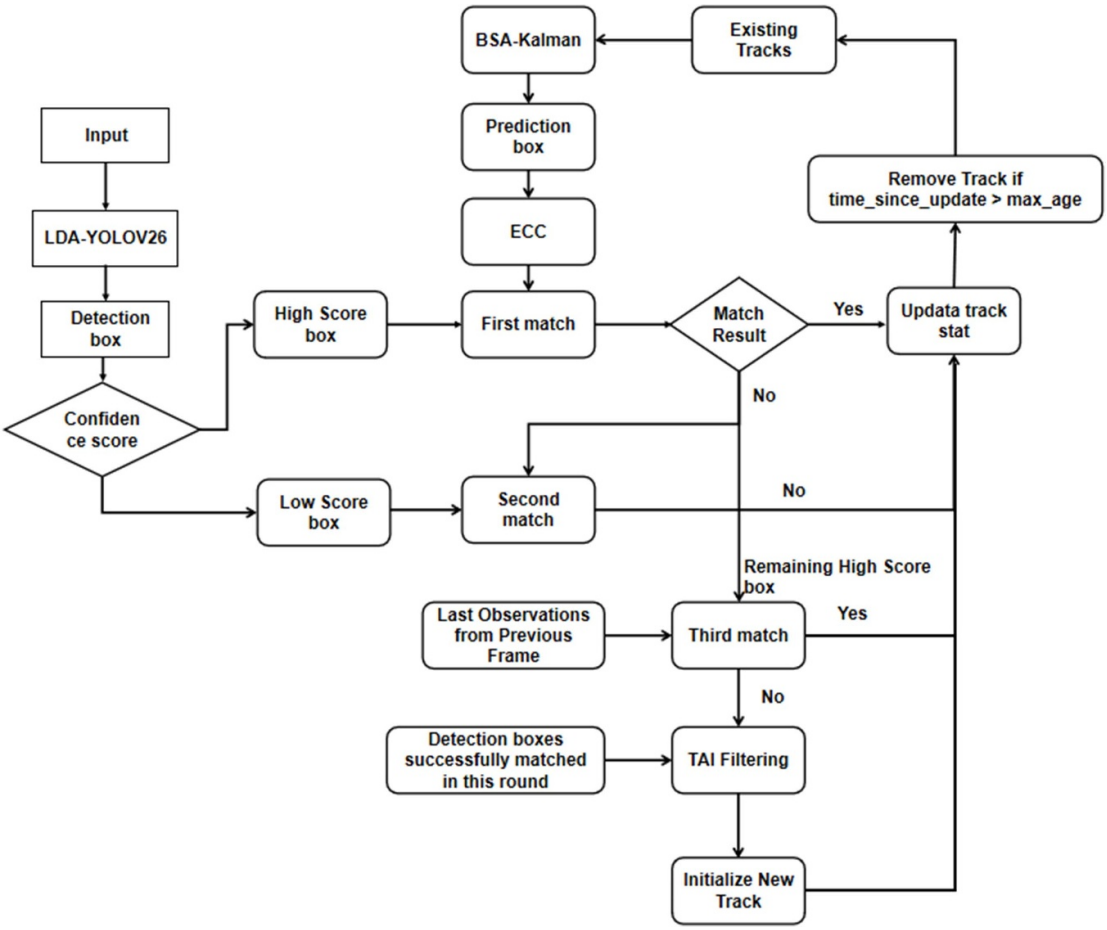


Figure 1. Overall framework of the proposed method.

driving tasks. The model is enhanced from the perspectives of feature fusion, global receptive field expansion, and spatial information interaction. The overall architecture of the improved LDA-YOLO26 model is shown in figures 2.

In the backbone stage, to address the large variation in target size at different distances, the LSK module is incorporated into the SPPF module to enlarge the receptive field, thereby enabling information extraction at multiple scales. In the Neck stage, an additional small-object detection head, P2, is introduced to enhance the model's ability to detect small objects that are distant, low-contrast, or partially occluded by the background. Meanwhile, DySample is adopted to replace the conventional UpSample operation, which improves the representation of fine-grained features and small objects in high-resolution feature maps while avoiding blurring and information loss. Furthermore, an ASFF-head detection head is designed to fuse features at different scales through an adaptive attention mechanism, effectively suppressing background interference and enhancing the response to target regions.

3.1.1. LSK-SPPF

In autonomous driving, vehicles are required to recognize objects at different scales. For example, nearby pedestrians and distant traffic signs may exhibit substantial size differences. To improve the model's capability to extract features across different scales, the LSK module is introduced into the SPPF module. The LSK-SPPF module is a feature enhancement unit constructed by incorporating a large-kernel separable attention mechanism into the original SPPF structure. Its overall pipeline consists of five stages: channel adjustment, multi-level pooling, feature concatenation, attention recalibration, and output mapping. Specifically, the input feature is first passed through a 1×1 convolution for channel compression, which reduces the number of channels and performs basic feature transformation while lowering the computational cost of subsequent operations. The compressed feature is then processed through three consecutive max-pooling operations, each configured with $k = 5$, $s = 1$, and $p = 2$. In this way, the receptive field is gradually enlarged without changing the spatial resolution, enabling the extraction of local contextual information at

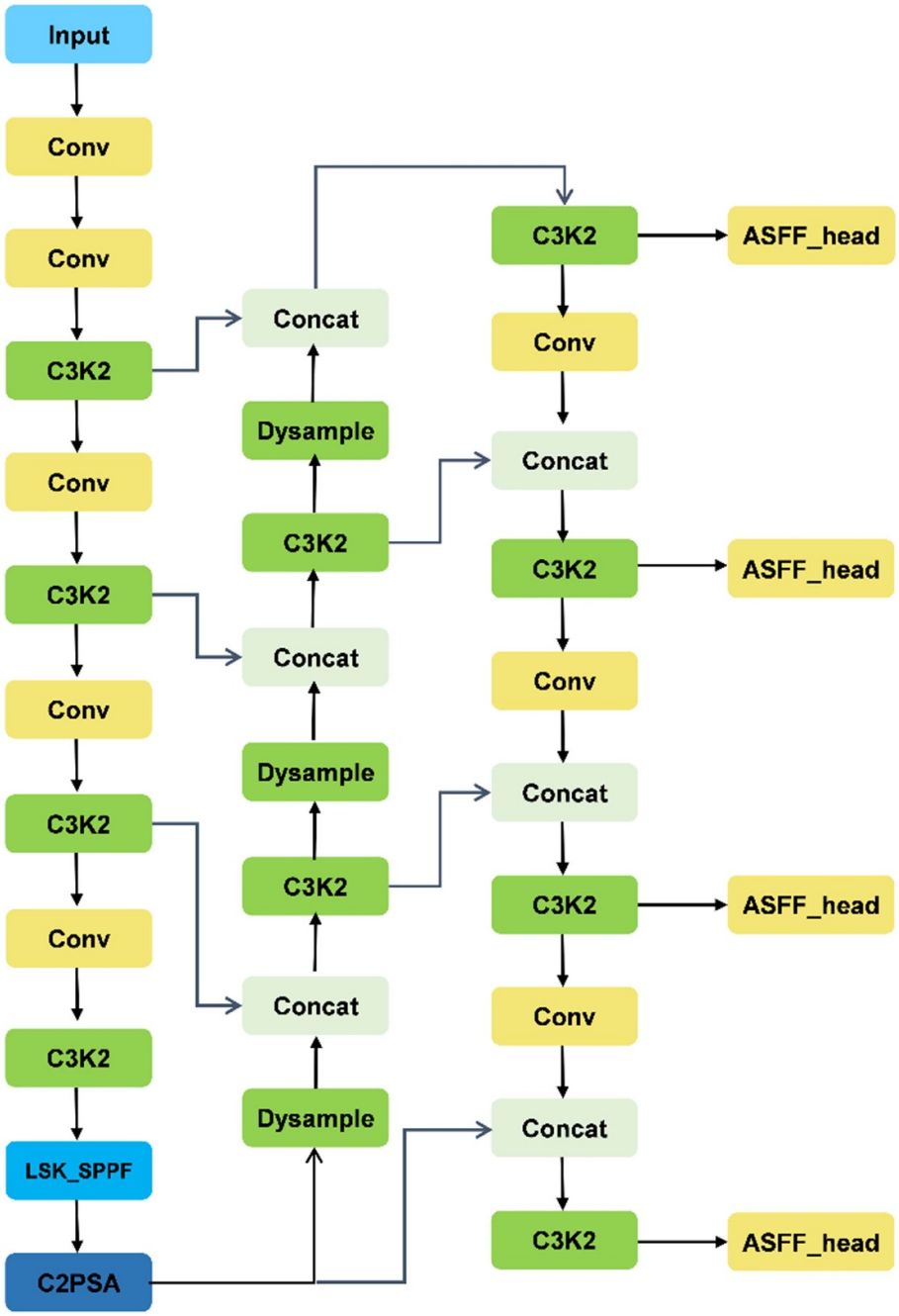


Figure 2. Network architecture of LDA-YOLO26.

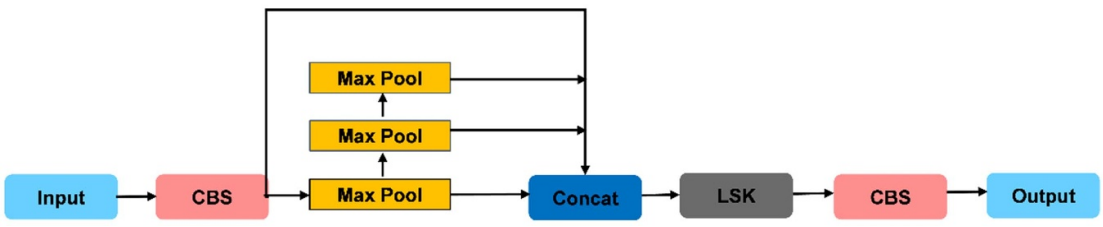


Figure 3. Network architecture of LSK-SPPF module.

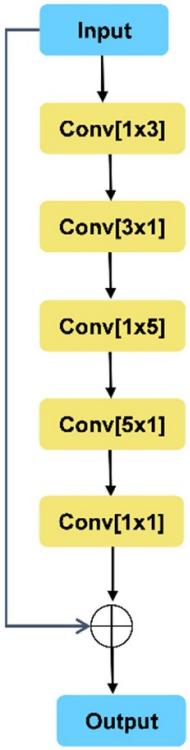


Figure 4. Network architecture of LSK module.

different levels. Subsequently, the original compressed feature, together with the first-, second-, and third-stage pooled features, is concatenated along the channel dimension to form a multi-scale fused representation.The LSK-SPPF network structure is shown in figures 3 and figures 4.

On this basis, an LSK attention branch is further introduced to perform spatial recalibration on the concatenated features. In the current implementation, the LSK module adopts a kernel size of $k = 11$ and models spatial dependencies sequentially through a set of horizontally and vertically separable convolutions. Specifically, 1×3 and 3×1 depthwise convolutions are first employed to extract basic directional information, followed by dilated 1×5 and 5×1 depthwise convolutions to further enlarge the effective receptive field. Finally, a 1×1 convolution is used to perform linear channel mapping and generate the corresponding attention response. The resulting attention map is then multiplied element-wise with the input feature, thereby adaptively enhancing key spatial regions and effective responses. After LSKA-based recalibration, the fused feature is mapped to the output channel c2 through another 1×1 convolution to produce the final output of the module. Overall, this structure combines the multi-scale contextual aggregation capability of SPPF with the large-kernel spatial selection capability of LSK, enabling the feature representation to achieve richer multi-scale expression and stronger spatial modeling ability while preserving the original resolution.

3.1.2. DSH-NECK

To further improve the representation capability of the feature fusion stage for multi-scale objects, two modifications are made to the Neck of the original network. Specifically, DySample is adopted to replace the conventional upsampling operator during feature scale restoration, and an additional small-object detection head, P2, is introduced into the detection stage to supplement the output branch for small-scale targets.

The added P2 detection branch performs prediction on a feature layer with higher spatial resolution, thereby providing finer positional cues and richer boundary details for small objects. Consequently, the model is able to achieve improved sensitivity and localization accuracy for small-scale targets in complex autonomous driving scenarios.

In the upsampling process, DySample first predicts sampling offsets through a 1×1 convolution and employs an additional branch to modulate the offset range. The predicted offsets are then combined with preset initial positions to generate dynamic sampling coordinates. Subsequently, feature resampling is performed through a differentiable grid sampling operation based on bilinear interpolation. In this way, the conventional fixed upsampling operation is transformed into a learnable dynamic resampling

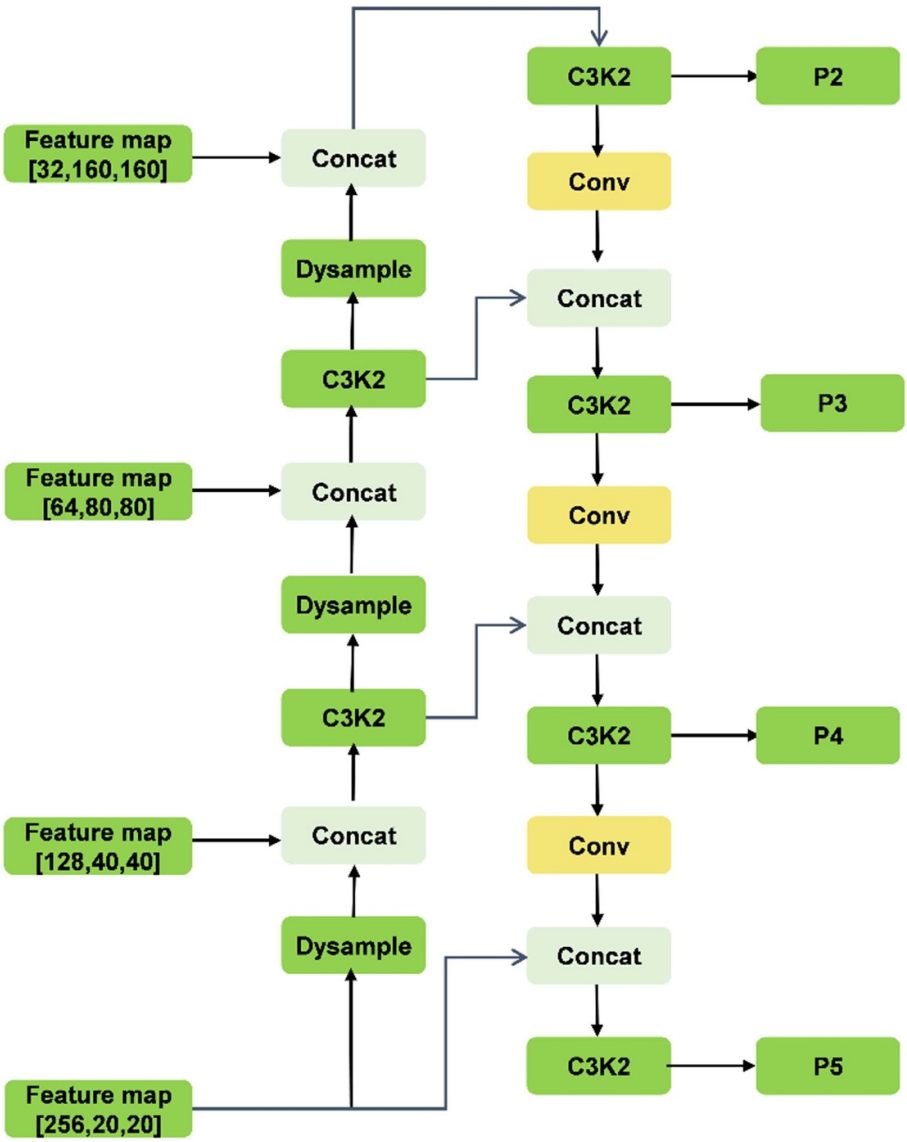


Figure 5. Network architecture of DSH-NECK.

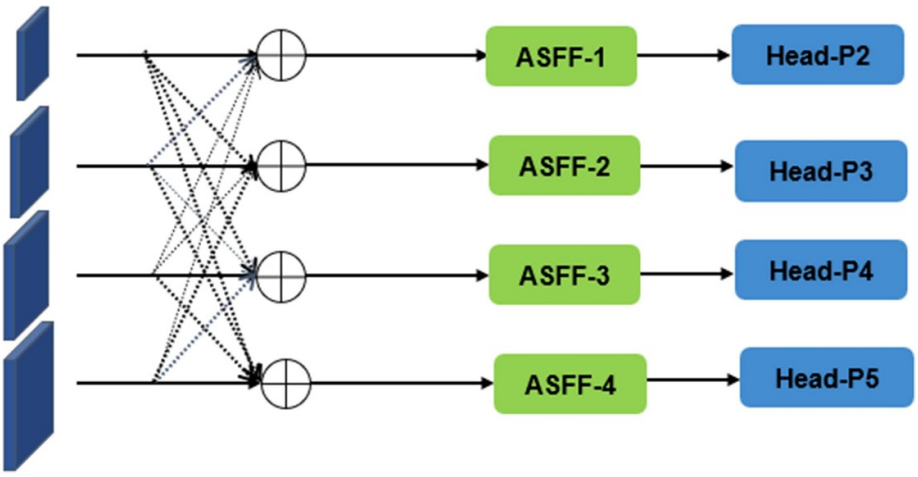


Figure 6. Network architecture of ASFF-HEAD.

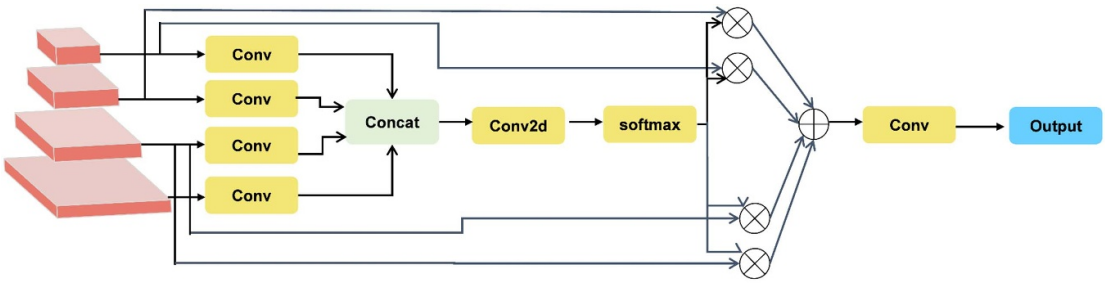


Figure 7. Network architecture of ASFF.

process, such that scale transformation is no longer limited to mere resolution expansion, but also incorporates a certain degree of feature selection and positional adjustment. This design is more conducive to adapting to the spatial distribution characteristics of features at different locations.The DSH-NECK network structure is shown in figures 5.The DSH-NECK network structure is shown in figures 5.

3.1.3. ASFF-HEAD

The ASFF module introduced in the Head enables the adaptive fusion of four feature maps. Its workflow can be summarized as follows. First, a 1×1 convolution is applied to each input branch to generate the corresponding weight representation. These representations are then concatenated and passed through another 1×1 convolution to produce the fusion weights. After softmax normalization, the four feature maps are combined through position-wise weighted summation, and the fused feature is finally output through a 3×3 convolution. This design allows the contribution of features at different scales to be dynamically selected according to spatial location, thereby achieving multi-scale information integration within a single feature layer and enhancing the interaction of multi-scale features.The ASFF-HEAD. network structure is shown in figures 6 and figures 7.

3.2. Experimental results and analysis

3.2.1. Dataset description

The KITTI dataset, proposed by Geiger *et al* [25], is a classical public dataset that has been widely used in the field of autonomous driving. It is commonly employed for tasks such as object detection, 3D object detection, depth estimation, and visual odometry. The dataset was collected in real-world road scenes using vehicle-mounted stereo cameras, LiDAR, and GPS/IMU sensors, covering complex traffic environments such as urban roads, rural roads, and highways. In this study, the detection subset of the KITTI dataset was used for the object detection task and was divided into a training set and a validation set at an approximate ratio of 8:2, including 5 985 training images and 1 496 validation images. The evaluated categories include three object classes: Car, Pedestrian, and Cyclist.

The BDD100K dataset, proposed by Yu *et al* [26], contains 100 000 driving video clips and covers various road scenarios, including urban streets, residential areas, and highways. In addition, it exhibits substantial diversity in weather conditions, illumination, and geographical regions, thereby reflecting the complex variations encountered in real traffic environments. For the generalization experiments, a subset of this dataset, namely BDD10K, was selected. The image resolution is 1280×720, and the dataset was divided into 7 000 training images, 1 000 validation images, and 2 000 test images.

3.2.2. Evaluation criteria

To objectively and quantitatively evaluate the performance of the proposed algorithm, this study mainly adopts Precision (P), Recall (R), and mean Average Precision (mAP) as evaluation metrics.

Precision refers to the proportion of correctly predicted positive samples among all samples predicted as positive, that is, the proportion of predicted targets that are truly relevant targets. Its calculation is given in equation (1). This metric is commonly used to evaluate whether the predictions of the network model are accurate and reliable,

$$P = \frac{TP}{TP+FP} \quad . \tag{1}$$

Recall refers to the proportion of correctly predicted positive samples among all actual positive samples. Its calculation is given in equation (2). This metric is mainly used to measure whether the network

Table 1. Definitions of variables in the evaluation metrics.

Variable	Definition
True positive (TP)	The number of actually positive targets that are correctly detected as positive samples
False positive (FP)	The number of actually negative targets that are incorrectly detected as positive samples
True negative (FN)	The number of actually positive targets that are incorrectly detected as negative samples

Table 2. Experimental environment and configuration.

Configuration item	Parameter
Operating system	Windows10
Deep learning framework	PyTorch 2.8.0
CUDA version	CUDA 12.9
Python version	Python3.10
CPU	Intel Core i9-9900K @ 3.60 GHz
GPU	NVIDIA RTX 4090

Table 3. Training parameters.

Item	Parameter
Cls_loss	BCE
Iou_loss	CIoU
Optimizer	MuSGD
Epochs	300
Batch size	32
Image size	640
Initial learning rate (Lr0)	0.01
Weight decay	0.0005
Momentum	0.937

model can accurately detect all targets during the detection process,

$$R = \frac{\mathrm{TP}}{\mathrm{TP+FN}} \tag{2}$$

mAP is a key metric for comprehensively evaluating network performance. It is mainly used to measure the average detection accuracy of the trained network model over all object categories. By integrating the precision values of different categories, it reflects the overall detection performance of the model across all classes. Its calculation is given in equation (3),

$$mAP = \frac{1}{N}\int_{\mathbb{R}} P(R)\,\mathrm{d}R \quad . \tag{3}$$

The meanings of the variables in the above equations are listed in table 1.

In addition, according to different IoU threshold criteria, mAP is further divided into two evaluation metrics, namely mAP@0.5 and mAP@0.5:0.95. Specifically, mAP@0.5 denotes the mAP value when the IoU threshold is set to 0.5, which mainly reflects the recognition capability of the algorithm. In contrast, mAP@0.5:0.95 represents the average mAP computed over IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05, which mainly reflects the algorithm's capability in object localization and bounding box regression.

3.2.3. Experimental environment and parameter settings

The experimental environment adopted in this study is summarized in table 2, and the training parameters are listed in table 3.

3.2.4. Ablation study and visualization analysis

To verify the effectiveness of each proposed improvement module in the object detection task, ablation experiments were conducted on the KITTI dataset using the baseline results of the YOLO26n model as the foundation. Each improved module was introduced individually to evaluate its specific contribution.

Table 4. Results of the ablation study on the KITTI dataset.

DSH-Neck	LSK-SPPF	ASFF-head	*p*	*r*	Map50	Map75	Map50: 95
			89.4%	82.8%	90.0%	73.5%	65.8%
√			89.9%	84.7%	91.6%	77.6%	68.4%
√	√		90.8%	84.9%	91.75	79.25	68.8%
√	√	√	**92.5%**	**87.4%**	**92.7%**	**81.7%**	**71.4%**

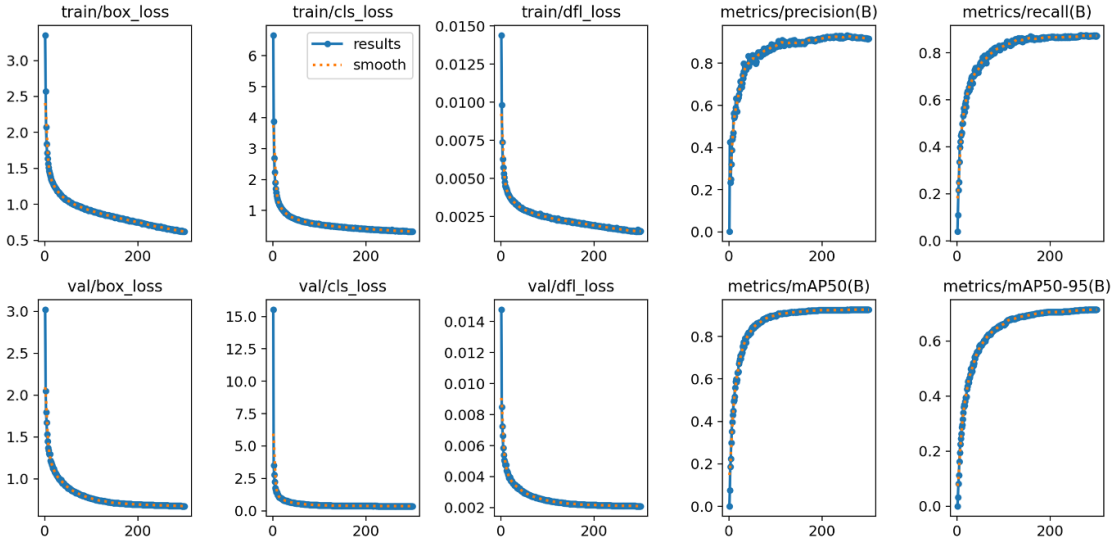


Figure 8. Curves of training metrics on the KITTI dataset.

The symbol '√' indicates that the corresponding module was incorporated. The experimental results are presented in table 4.(The bold marks show the results we achieved)

As can be seen from table 4, after the DSH-Neck module was introduced into the YOLO26 baseline model, the Precision, Recall, mAP@0.5, mAP@0.75, and mAP@0.5:0.95 increased from 89.4%, 82.8%, 90.0%, 73.5%, and 65.8% to 89.9%, 84.7%, 91.6%, 77.6%, and 68.4%, respectively, corresponding to improvements of 0.5%, 1.9%, 1.6%, 4.1%, and 2.6%. These results indicate that the module can effectively enhance the feature fusion capability of the neck network and improve the detection performance of the model for road-scene targets in the KITTI dataset.

Furthermore, after the LSK-SPPF module was added on this basis, Precision, Recall, mAP@0.5, mAP@0.75, and mAP@0.5:0.95 further increased to 90.8%, 84.9%, 91.75%, 79.25%, and 68.8%, respectively, representing improvements of 1.4%, 2.1%, 1.75%, 5.75%, and 3.0% over the baseline. This demonstrates that the module can further enlarge the receptive field and enhance the model's ability to extract multi-scale target features.

In addition, after integrating the ASFF-head detection head, all evaluation metrics were further improved to 92.5%, 87.4%, 92.7%, 81.7%, and 71.4%, respectively, which correspond to gains of 3.1%, 4.6%, 2.7%, 8.2%, and 5.6% over the baseline. These results show that the module can achieve more efficient adaptive fusion of features at different scales, thereby significantly enhancing the model's detection capability for multi-scale targets in complex scenes, especially small and distant objects.

It can also be observed from the table that, when used in combination, all proposed modules yield further improvements in each evaluation metric compared with the single-module schemes. This verifies the effectiveness and compatibility of the proposed improvements and demonstrates that they can effectively enhance the overall detection performance of YOLO26 on the KITTI dataset.

As shown in figure 8, the training curves of the proposed model on the KITTI dataset demonstrate a stable convergence process. The training losses, including box loss, classification loss, and DFL loss, decrease rapidly in the early training stage and then gradually converge, indicating that the model can effectively learn object localization and classification features. Meanwhile, the validation losses show a similar downward trend and remain stable in the later stage, suggesting that the model has good generalization ability and does not exhibit obvious overfitting.

In terms of detection performance, the precision, recall, mAP@0.5, and mAP@0.5:0.95 curves increase steadily with the training epochs and eventually tend to stabilize. This indicates that the

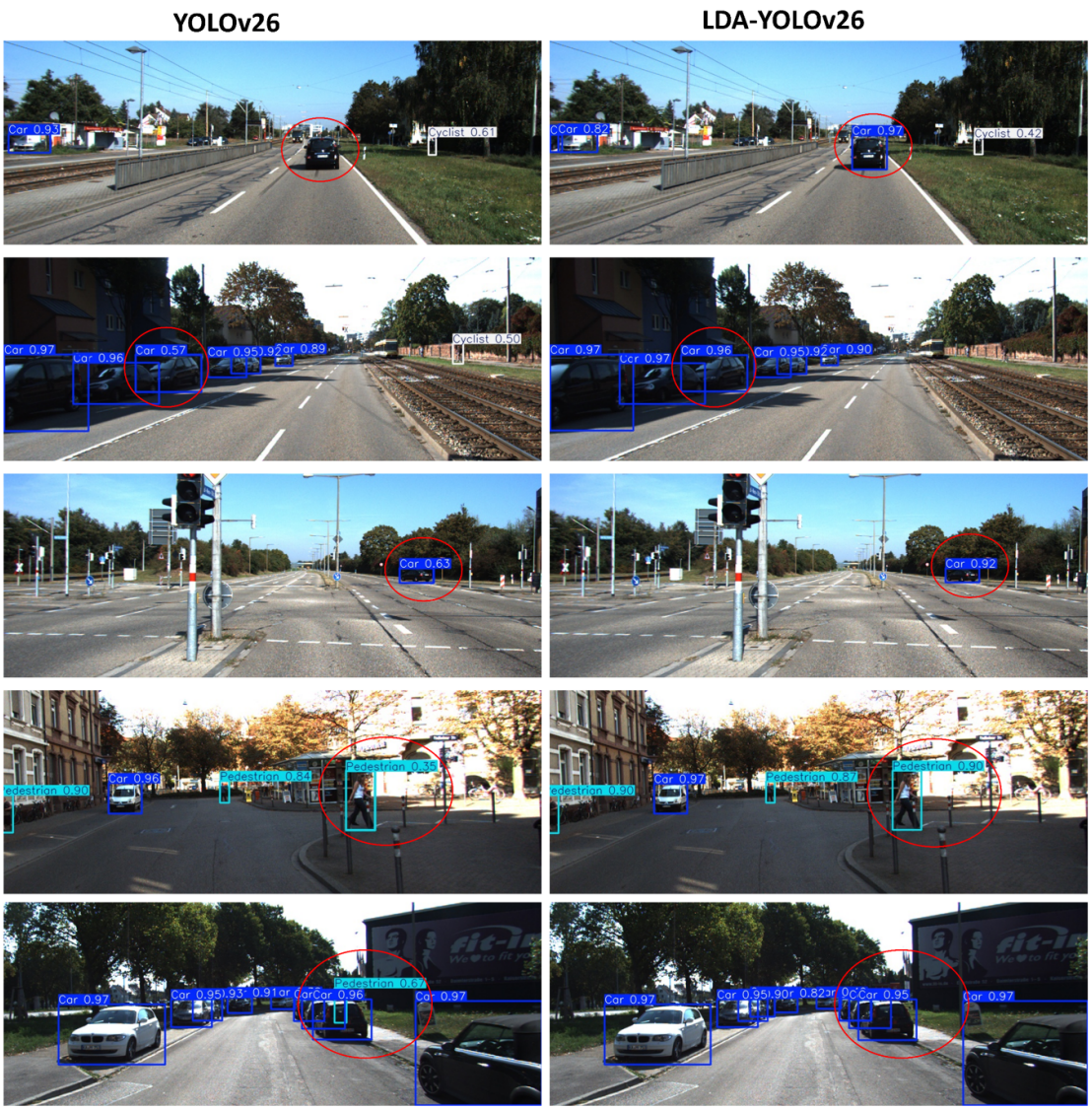


Figure 9. Visualization comparison of results on the KITTI dataset.

proposed model gradually improves its detection accuracy and target recognition capability during training. Overall, the results in figure 8 verify the effectiveness and convergence stability of the improved YOLO26 model on the KITTI dataset.

To further illustrate the effectiveness of the improved model, a visual analysis of the experimental results on the KITTI dataset was conducted, as shown in figure 9.

As shown in figure 9, the improved model consistently outperforms the original YOLO26 in detection performance under both daytime and nighttime scenarios on the BDD100K dataset. For key traffic participants, such as vehicles and pedestrians, the enhanced model achieves more complete and accurate target recognition and localization, with particularly notable advantages in detecting small-scale, long-distance, and partially occluded objects. As highlighted by the red-marked regions, the baseline model is more prone to missed detections and false positives in complex backgrounds, low-illumination conditions, and densely populated scenes, whereas the improved model effectively alleviates these issues and exhibits stronger perception capability for weak and easily confused targets. In addition, the improved model demonstrates greater stability in terms of bounding-box localization and category discrimination. These qualitative comparison results indicate that the proposed improvement strategy effectively enhances the feature extraction and multi-scale representation capabilities of the model, improves its adaptability to complex traffic environments, and further confirms the effectiveness and practical value of the proposed method for object detection in autonomous driving scenarios.

Table 5. Experimental results on the BDD100K dataset.

	p	*r*	Map50	Map75	Map50-95
YOLO26	54.8%	33.3%	34.5%	15.8%	18.0%
LDA-YOLO26	60.6%	36.4%	38.7%	19.7%	21.4%

3.2.5. Generalization experiments and visualization analysis

To verify the generalization capability of the improved model, generalization experiments were conducted on the BDD100K dataset. The experimental results are presented in table 5.

On the BDD100K dataset, LDA-YOLO26 demonstrates a clear overall performance improvement over YOLO26. Specifically, Precision, Recall, mAP@0.5, mAP@0.75, and mAP@0.5:0.95 are improved by 5.8%, 3.1%, 4.2%, 3.9%, and 3.4%, respectively. These results indicate that the proposed method can effectively enhance the detection accuracy and target recall capability of the model, while also achieving favorable performance across different scenarios.

As shown in the visual comparison between YOLO26 and LDA-YOLO26 on the BDD100K dataset in figure 10, LDA-YOLO26 demonstrates superior detection performance in complex road scenes. Compared with the original YOLO26, the improved model shows greater effectiveness in detecting small, edge-located, and occluded objects. Specifically, the first row indicates that LDA-YOLO26 achieves more accurate category classification and bounding box regression for vehicle targets. The results in the second and third rows show that, under low-light nighttime conditions, the improved model can more effectively detect distant vehicles, traffic signs, and pedestrians, with fewer missed detections. The fourth and fifth rows further demonstrate that, in challenging scenarios such as snowy weather and crowded urban roads, LDA-YOLO26 still maintains strong detection capability for distant pedestrians and occluded targets.

3.2.6. Comparative experiments

To verify the superiority of LDA-YOLO26, comparative experiments were conducted with mainstream object detection algorithms. The comparison results on the KITTI dataset are presented in table 6.

(Given that the experimental hardware and evaluation settings for the models reported in [31.32.33.34] are highly similar to those in this study, it is reasonable and valid to directly cite their original data for comparison.).

To verify the superiority of the proposed LDA-YOLO26 model, comparative experiments were conducted on the KITTI dataset against several mainstream object detection algorithms, and the results are presented in table 6. Overall, LDA-YOLO26 demonstrates strong competitiveness across all evaluation metrics. Specifically, LDA-YOLO26 achieves a precision of 92.5%, a recall of 87.4%, an mAP@0.5 of 92.7%, and an mAP@0.5:0.95 of 71.4%. Among these metrics, both mAP@0.5 and mAP@0.5:0.95 surpass those of Faster R-CNN, YOLOv8s, YOLOv9s, YOLOv10s, YOLOv11s, and YOLOv12s, indicating that the proposed method has a clear advantage in detection accuracy. In particular, it maintains strong performance under the more stringent multi-threshold evaluation criterion, reflecting its superior localization capability and overall detection robustness.

From the perspective of computational complexity, the FLOPs of LDA-YOLO26 are only 10.2 G, which is significantly lower than those of the other compared models. For example, compared with YOLOv8s, YOLOv9s, YOLOv10s, and YOLOv11s, the proposed method substantially reduces computational cost while maintaining or even improving detection accuracy. In comparison with Faster R-CNN, it achieves higher detection accuracy with a much lower computational burden. These results indicate that LDA-YOLO26 has higher computational efficiency in network design and is capable of achieving better detection performance with fewer computational resources. Although its precision is slightly lower than that of some models, such as YOLOv11s, it achieves superior results in recall, mAP@0.5, and mAP@0.5:0.95, which more comprehensively reflect overall detection performance. Therefore, its overall performance remains the best among the compared methods.

In terms of inference speed, LDA-YOLO26 achieves 463 FPS, which satisfies the requirement of real-time object detection. Its FPS is higher than those of YOLOv8s, YOLOv9s, YOLOv11s, YOLOv12s, YOLOv8n, YOLOv11n, and YOLOv12n. Although YOLOv10n achieves the highest FPS, its detection accuracy is much lower than that of LDA-YOLO26, especially in mAP@0.5:0.95, where it is 8.6% lower. Compared with YOLOv10s, LDA-YOLO26 has only a slight decrease in FPS, but achieves higher recall, mAP@0.5, and mAP@0.5:0.95 with much lower FLOPs. Therefore, LDA-YOLO26 does not simply pursue lightweight design or inference speed, but achieves a more favorable trade-off among detection accuracy, computational cost, and real-time performance.



Figure 10. Visual comparison results on the BDD100K dataset.

4. Improved OCSORT-based object tracking algorithm

4.1. Overall framework

To address challenges encountered by tracking methods in autonomous driving scenarios, such as target disappearance and trajectory drift, this study takes OCSORT as the baseline model and further optimizes it, thereby proposing an improved model termed BET-OCSORT.

The improved OCSORT matching pipeline retains the advantages of the original two-stage association mechanism while further integrating the enhanced detection model, LDA-YOLO26, to establish a more comprehensive three-stage association strategy. First, to address the state estimation bias that frequently arises in conventional trajectory prediction under complex motion scenarios, BSA-Kalman is introduced into the trajectory update process for motion prediction. After the predicted bounding boxes are obtained, ECC-based camera motion compensation is further applied to refine the prediction results. This effectively alleviates the positional deviations caused by camera shake and viewpoint

Table 6. Comparative experimental results on the KITTI dataset.

Model	p	r	map@0.5	map@0.5:0.95	Flops	FPS
Faster RCNN [27]	84.5%	77.7%	80.9%	67.5%	60.5 G	—
RT-DETR-I [28]	81.7%	80.2%	86.8%	—	—	—
SSD300 [29]	91.3%	82.1%	90.9%	—	60.9 G	—
RetinaNet(ResNet50) [30]	87.4%	75.8%	78.8%	48.7%	—	—
YOLOv8s	92.8%	86.7%	92.3%	68. 9%	28.4 G	350
YOLOv8n	88.5%	81.5%	88.2%	63.6%	8.1 G	337
YOLOv9s	93.2%	85.9%	82.2%	68.8%	26.7 G	325
YOLOv10s	91.7%	85.1%	91.4%	68.4%	21.4 G	485
YOLOv10n	88.4%	80.1%	87.3%	62.8%	6.5 G	645
YOLOv11s	94.1%	85.1%	91.8%	68.5%	21.3 G	339
YOLOv11n	91.8%	80.0%	88.1%	63.1%	6.3 G	437
YOLOv12s	89.6%	83.9%	89.5%	63.7%	21.2 G	272
YOLOv12n	86.3%	76.7%	84.5%	57.6%	6.3 G	354
YOLO26n	89.4%	82.8%	90.0%	65.8%	6.3 G	480
LDA-YOLO26(ours)	92.5%	87.4%	92.7%	71.4%	10.2 G	463

Table 7. Definitions of the parameters in the equations.

Model	Flops
p	Geometric transformation parameters
$\rho(p)$	Enhanced correlation coefficient
$T(x)$	Template image, typically the previous frame
$I(x)$	Input image, typically the current frame
$W(x;p)$	Geometric transformation parameterized by p
$\bar{T}$	Mean intensity of the template image
$\bar{I}_p$	Mean intensity of the warped input image

Table 8. Definitions of the variables used in the evaluation metrics.

Variable	Definition
FP	False positives
TP	Correctly matched detections
FN	Missed detections
GT_{total}	Total number of ground-truth objects across all frames
IDTP	Detections with correctly matched identities
IDFP	Number of false positives at the identity level
ICFN	Number of false negatives at the identity level

Table 9. Ablation results on the KITTI dataset.

ECC	BSA-KF	TAI	MOTA	HOTA	IDF1	IDSW	FPS
			88.97%	74.78%	79.03%	251	35.6
√			90.07%	78.24%	83.42%	198	23.2
√	√		91.55%	80.09%	86.53%	184	21.1
√	√	√	92.28%	83.69%	91.73%	99	21.2

variation under moving-camera conditions, thereby improving the consistency between the predicted boxes and the true target locations. Second, in the first association stage, high-confidence detection boxes are matched with the motion-compensated predicted trajectories, giving priority to the association and update of high-quality targets and thus ensuring the stability of the primary trajectories. For trajectories that fail to be matched in the first stage, the framework preserves the second association mechanism with low-confidence detection boxes, allowing low-score detections to recover some missed targets when confidence decreases due to occlusion, blur, or long-distance observation, and thereby enhancing the utilization of weak and marginal targets. Finally, a third association stage is performed between the detection results from the previous frame and the remaining high-confidence detections. This enables compensatory association through historical observations in the presence of short-term occlusion, abrupt

motion changes, or fluctuations in detection results, thereby effectively reducing trajectory fragmentation. Before initializing new trajectories, the TAI mechanism is further employed to filter candidate targets, which effectively suppresses the direct generation of false trajectories from false-positive detections. Through the collaborative design of BSA-Kalman prediction, ECC compensation, three-stage association, and TAI filtering, the improved matching pipeline significantly enhances the robustness of target association and the stability of tracking in complex traffic scenarios, dynamic camera-view environments, and heavily occluded conditions, thus providing effective support for improving vehicle multi-object tracking performance.

4.1.1. BSA-KF

In autonomous driving target tracking, the most common objects are vehicles and pedestrians, whose actual motion patterns are usually complex. To better handle variations in target scale and motion state, the Kalman filter in the original OCSORT algorithm is improved.

The improved state vector of the Kalman filter is defined as follows:

$$x = \begin{bmatrix} c_x & c_y & w & h & v_x & v_y & v_w & v_h \end{bmatrix}^T \tag{4}$$

c_x, c_y: center coordinates of the target
w, h;: width and height of the target
v_x, v_y: velocities of the target center
v_w, v_h: variation velocities of the target width and height.

Compared with the original formulation, the improved state vector directly estimates the target width and height, enabling the tracker to capture variations in bounding-box scale. This design is more suitable for multi-object tracking scenarios in which significant target size changes frequently occur.

In conventional target tracking, fixed process noise covariance matrix Q and observation noise covariance matrix R are usually adopted, which makes it difficult to characterize the influence of target scale variation on state estimation. Therefore, while preserving the basic recursive framework of the Kalman filter, the process noise covariance matrix Q is dynamically constructed according to the target width W and height H in the current state. The improved formulation is given in equation (5):

$$Q = \mathrm{diag}\left((\sigma_{\mathrm{p}} w)^2, (\sigma_{\mathrm{p}} h)^2, (\sigma_{\mathrm{v}} w)^2, (\sigma_{\mathrm{v}} h)^2, (\sigma_{\mathrm{v}} w)^2, (\sigma_{\mathrm{v}} h)^2 \right) \tag{5}$$

w, h: width and height in the current state estimation
σ_{p}: proportional factor of the positional process noise
σ_{v}: proportional factor of the velocity process noise.

Similarly, in the update stage, the observation noise covariance matrix R is dynamically generated according to the width and height information of the observed bounding box or the current state:

$$R = \mathrm{diag}\left((\sigma_r w)^2, (\sigma_r h)^2, (\sigma_r w)^2, (\sigma_r h)^2 \right) \tag{6}$$

w, h: width and height of the measured bounding box
σ_r: proportional factor of the measurement noise.

It should be noted that BSA-KF shares the general motivation of adaptive Kalman filtering with existing methods, such as NSA-Kalman and confidence-/scale-adaptive noise mechanisms used in recent trackers. However, the proposed BSA-KF differs from these methods in terms of adaptive factors, covariance design, and application motivation. NSA-Kalman mainly adjusts the observation noise according to the detection confidence score, where the reliability of the measurement is determined by the detector output confidence. In contrast, BSA-KF constructs the noise covariance according to the geometric scale of the target bounding box, namely its width and height. This design is motivated by autonomous driving scenarios, where vehicles, pedestrians, and cyclists often undergo significant scale changes caused by perspective variation, distance changes, occlusion, and camera motion. Moreover, BSA-KF dynamically adjusts both the process noise covariance matrix (Q) and the observation noise covariance matrix (R), rather than only modulating the measurement noise. The improved state vector explicitly includes the target width, height, and their variation velocities, enabling the tracker to model bounding-box scale evolution directly. Therefore, BSA-KF should be regarded as a bounding-box-scale-aware adaptive covariance design integrated into the improved OCSORT framework, rather than a simple reproduction of existing confidence-adaptive Kalman filtering methods. By incorporating scale-aware (Q) and (R) estimation into the prediction and update process, BSA-KF improves the robustness of trajectory estimation in complex autonomous driving scenes with scale variation, occlusion, and moving-camera interference.

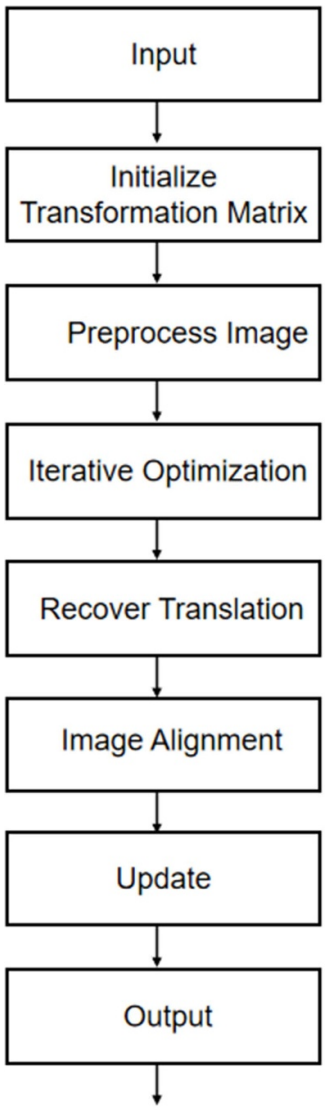


Figure 11. Workflow of the ECC algorithm.

4.1.2. ECC-based camera motion compensation

During vehicle motion, the camera pose changes continuously because of the vehicle's dynamic characteristics, often causing translation and rotation between consecutive frames. Consequently, the captured vehicles may undergo positional displacement and shape distortion. To alleviate the adverse effects of camera shake and viewpoint changes on multi-object tracking, this study adopts an ECC-based camera motion compensation method. The overall procedure is shown in figure 11.

Formula for the ECC correlation coefficient:

$$\mathbf{p}^{*} = \arg\max_{\mathbf{p}} \rho(p) \tag{7}$$

$$\rho(p) = \frac{\sum_{x}\left(T(x)-\bar{T}\right)\left(I(W(x;\mathbf{p}))-\bar{I}_{\mathbf{p}}\right)}{\sqrt{\sum_{x}\left(T(x)-\bar{T}\right)^{2}\sum_{x}\left(I(W(x;\mathbf{p}))-\bar{I}_{\mathbf{p}}\right)^{2}}} \quad . \tag{8}$$

The fundamental idea of the ECC method is to construct a parameterized geometric transformation model between two consecutive frames and iteratively adjust the transformation parameters, including translation, rotation, affine transformation, and homography, to find the optimal solution that maximizes the correlation coefficient. As a result, one frame can be geometrically transformed to align with the other frame as closely as possible in the spatial domain. The definitions and units of all parameters are summarized in Table 7.

4.1.3. Track-aware initialization (TAI)

The OCSORT method typically initializes all unmatched detection boxes with sufficiently high confidence scores as new trajectories. However, these detection boxes may include candidates that highly overlap with currently active trajectories, or redundant candidates caused by occlusion, noise, reflections, and other disturbances. To suppress the generation of false and duplicate trajectories, the TAI strategy is introduced. Specifically, the final positions of the matched trajectories after two rounds of high-confidence association are treated as non-removable anchors. These anchors are then jointly processed with the unmatched high-confidence detection boxes using NMS. After suppression, only the remaining detection boxes are allowed to initialize new trajectories. This strategy effectively reduces false trajectories, thereby improving the quality of trajectory initialization and enhancing the stability of subsequent association in online multi-object tracking.

4.2. Experimental results and analysis

4.2.1. Dataset description

In this section, the official TrackEval evaluation toolkit was used to evaluate 21 video sequences from the KITTI dataset and 200 video sequences from the BDD100K dataset. Only the vehicle category was considered in the evaluation.

4.2.2. Evaluation metrics

To comprehensively and objectively evaluate the performance of the multi-object tracking algorithm, this study adopts HOTA (higher order tracking accuracy), MOTA (multiple object tracking accuracy), IDF1 (Identification F1 Score), and IDSW as the primary evaluation metrics. The definitions and units of all parameters are summarized in Table 8.

MOTA: MOTA is used to measure the overall accuracy of a multi-object tracking algorithm by jointly considering three major types of errors, namely missed detections, false positives, and identity switches. It is calculated as follows:

$$\mathrm{MOTA} = 1 - \frac{\mathrm{FN}+\mathrm{FP}+\mathrm{IDSW}}{\mathrm{GT}_{\mathrm{total}}} \quad . \tag{9}$$

A higher MOTA value indicates better overall tracking performance. However, this metric is relatively sensitive to detection results and mainly reflects the quality of detection and the overall error level.

IDF1: IDF1 is mainly used to measure the accuracy of identity preservation during tracking, and it reflects the ability of an algorithm to maintain identity consistency over long temporal sequences. It is calculated as follows:

$$\mathrm{IDF1} = \frac{2\cdot\mathrm{IDTP}}{2\cdot\mathrm{IDTP}+\mathrm{IDFP}+\mathrm{IDFN}} \quad . \tag{10}$$

HOTA: HOTA provides a balanced evaluation of detection accuracy and association accuracy, and more comprehensively reflects the overall performance of a tracker in terms of detection, association, and localization. It is calculated as follows:

$$\mathrm{DetA}_{\alpha} = \frac{|\mathrm{TP}|}{|\mathrm{TP}|+|\mathrm{FN}|+|\mathrm{FP}|} \tag{11}$$

$$\mathrm{HOTA}_{\alpha} = \sqrt{\mathrm{DetA}_{\alpha}\cdot\mathrm{AssA}_{\alpha}}. \tag{12}$$

IDSW: IDSW denotes the total number of identity switches that occur across all frames.

4.2.3. Ablation experiments and visualization analysis

To verify the effectiveness of each improved module in the tracking task, ablation experiments were conducted on the KITTI dataset in this section, using LDA-YOLO26 as the feature extraction network.The ablation results for different combinations of the proposed modules are summarized in Table 9.

The experimental results show that, compared with the baseline, the progressive incorporation of ECC, BSA-KF, and TAI on the KITTI dataset led to continuous improvements in all tracking evaluation metrics. When only the ECC module was introduced, MOTA increased from 88.97% to 90.07%, representing an improvement of 1.10%; HOTA increased from 74.78% to 78.24%, representing an improvement of 3.46%; and IDF1 increased from 79.03% to 83.42%, representing an improvement of 4.39%. Meanwhile, IDSW decreased from 251 to 198, corresponding to a reduction of 53. These results indicate that the ECC module can effectively improve target matching quality and reduce identity switches during the tracking process.

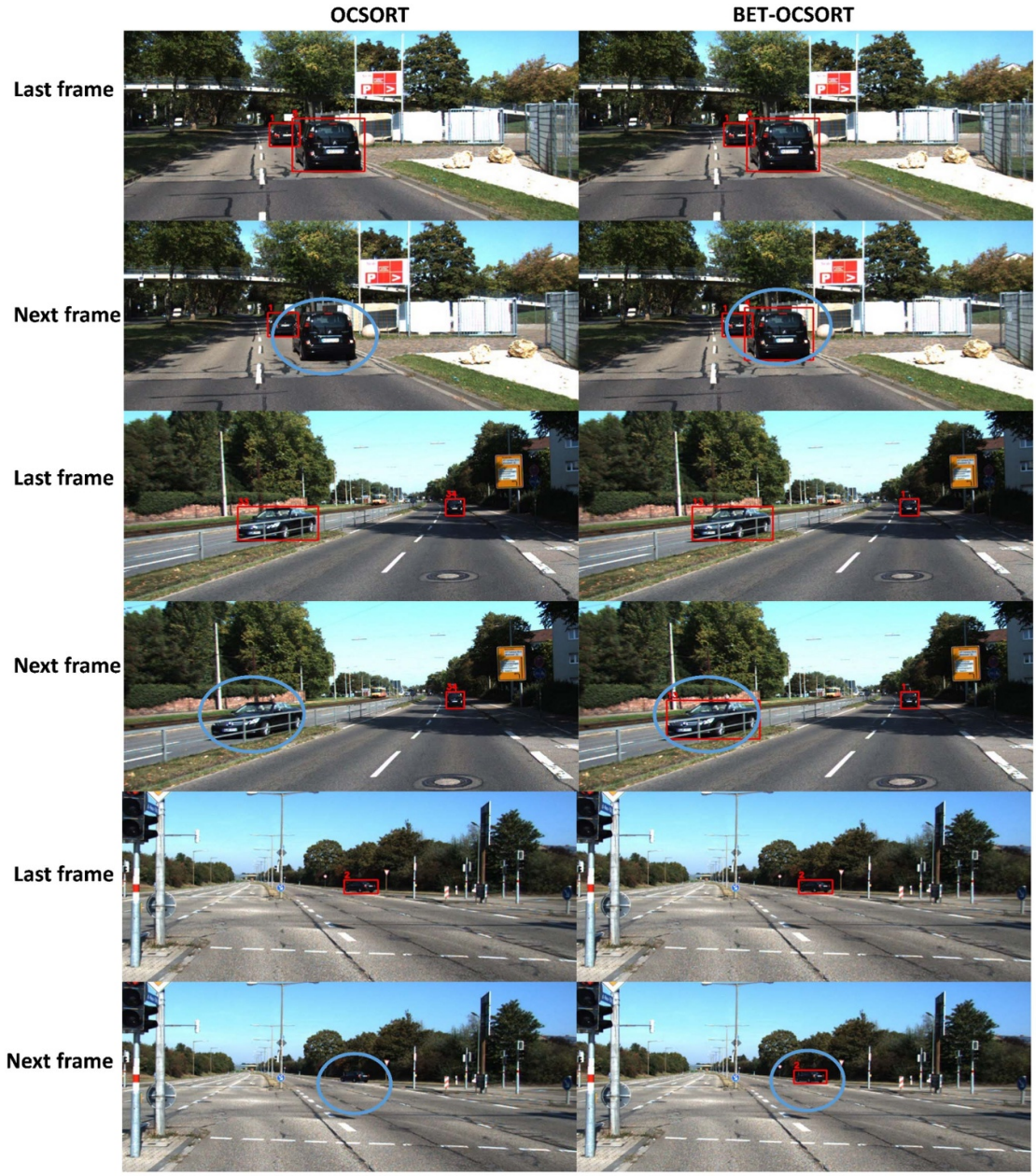


Figure 12. Visualization of tracking results on the KITTI dataset.

After further introducing BSA-KF, MOTA increased to 91.55%, HOTA increased to 80.09%, IDF1 increased to 86.53%, and IDSW decreased to 184. These results demonstrate that the BSA-KF module can further enhance the stability of target state estimation and trajectory prediction, thereby contributing positively to the overall model performance.

Finally, after incorporating TAI on top of ECC and BSA-KF, the model achieved the most significant performance improvement. Specifically, MOTA reached 92.28%, HOTA reached 83.69%, representing an overall improvement of 8.91%, IDF1 reached 91.73%, and IDSW decreased to 99. These results indicate that TAI plays a significant role in improving identity preservation, reducing identity switches, and enhancing the consistency of trajectory association.

Although the FPS decreases to some extent, the additional computational overhead is acceptable considering the significant improvements in tracking accuracy, identity preservation, and overall robustness.

Overall, compared with the baseline method, the improved tracking method achieved substantial gains in MOTA, HOTA, and IDF1, while IDSW was significantly reduced. This demonstrates that all the proposed modules contribute positively to the vehicle multi-object tracking task.

To further demonstrate the effectiveness of the improved model, a visualization analysis of the tracking results on the KITTI dataset was conducted, as shown in figure 12. Compared with the original

Table 10. Ablation results on the BDD100K dataset.

ECC	BSA-KF	TAI	MOTA	HOTA	IDF1	IDSW
			41.20%	41.40%	46.61%	10 101
√			42.21%	43.89%	49.38%	9937
√	√		44.00%	44.02%	50.15%	8631
√	√	√	44.10%	47.20%	53.72%	8256

model, the proposed method exhibits stronger data association continuity under occlusion and target motion scenarios, effectively reducing identity switches and improving tracking stability.

4.2.4. Generalization experiments and visualization analysis

To further verify the generalization capability of the improved model, the ablation results on the BDD100K dataset are presented in table 10.

The ablation results on the BDD100K dataset show that the overall tracking performance of the model was consistently improved with the progressive introduction of the ECC, BSA-KF, and TAI modules. After incorporating ECC, MOTA, HOTA, and IDF1 increased to 42.21%, 43.89%, and 49.38%, respectively, while IDSW decreased to 9 937, indicating that this module contributes to target alignment and association in complex scenes. On this basis, after further introducing BSA-KF, MOTA increased to 44.00%, and IDSW was significantly reduced to 8 631, indicating that this module is effective in motion state modeling and trajectory prediction. After further incorporating TAI, HOTA and IDF1 increased to 47.20% and 53.72%, respectively, while IDSW was further reduced to 8 256, demonstrating that this module plays a more prominent role in enhancing identity preservation and improving the consistency of trajectory association. Overall, all the proposed modules make positive contributions to the improvement of multi-object tracking performance on the BDD100K dataset, which validates the generalization capability of the model and demonstrates its superiority across different scenarios.

The visualization results are shown in figure 13. As illustrated in the figure, the improved model achieves substantially better vehicle tracking performance in urban street scenes and effectively alleviates trajectory fragmentation in distant and occluded scenarios.

4.2.5. Comparative experiments

To verify the superiority of the improved method in similar tasks, comparative experiments were conducted on the KITTI dataset. To ensure fairness, all tracking algorithms included in the comparison used the improved LDA-YOLO26 model as the detector. The comparison results on the KITTI dataset are presented in table 11.

The experimental results show that the proposed method outperforms most of the compared tracking algorithms listed in the table on the current dataset, and demonstrates clear advantages in terms of overall tracking performance and identity association stability. The proposed method achieves the highest HOTA value among all compared methods, surpassing the other methods by 3.68%, 3.17%, 16.94%, 29.84%, and 2.69%, respectively, indicating that it attains a better balance between detection accuracy and data association quality. Meanwhile, the proposed method yields the lowest IDSW among all methods, with reductions of 46, 253, 199, 542, and 78, respectively, compared with the aforementioned methods. This demonstrates that the proposed method can more effectively suppress identity switches during tracking and exhibits stronger identity preservation capability in complex scenarios involving occlusion, target interaction, and appearance variation.

Although the proposed method is 0.88 percentage points lower than StrongSORT in terms of MOTA and 1.81 percentage points lower than ByteTrack in terms of IDF1, the IDSW values of StrongSORT and ByteTrack reach 352 and 177, respectively, which are significantly higher than that of the proposed method. This indicates that, although these two methods perform better on certain individual metrics, they still exhibit limitations in identity consistency and long-term association stability. In contrast, the proposed method achieves the highest HOTA and the lowest IDSW while maintaining relatively high MOTA and IDF1, demonstrating stronger global association robustness and more balanced overall tracking performance.

ByteTrack adopts a more recall-oriented association strategy, which tends to preserve low-confidence detection results more sufficiently. Therefore, it may obtain higher identity matching coverage and thus achieve a slightly higher IDF1 score. In contrast, while the proposed method also utilizes low-confidence detection boxes, it further incorporates BSA-Kalman prediction, ECC-based camera motion compensation, three-stage association, and TAI filtering into the overall matching pipeline. These mechanisms

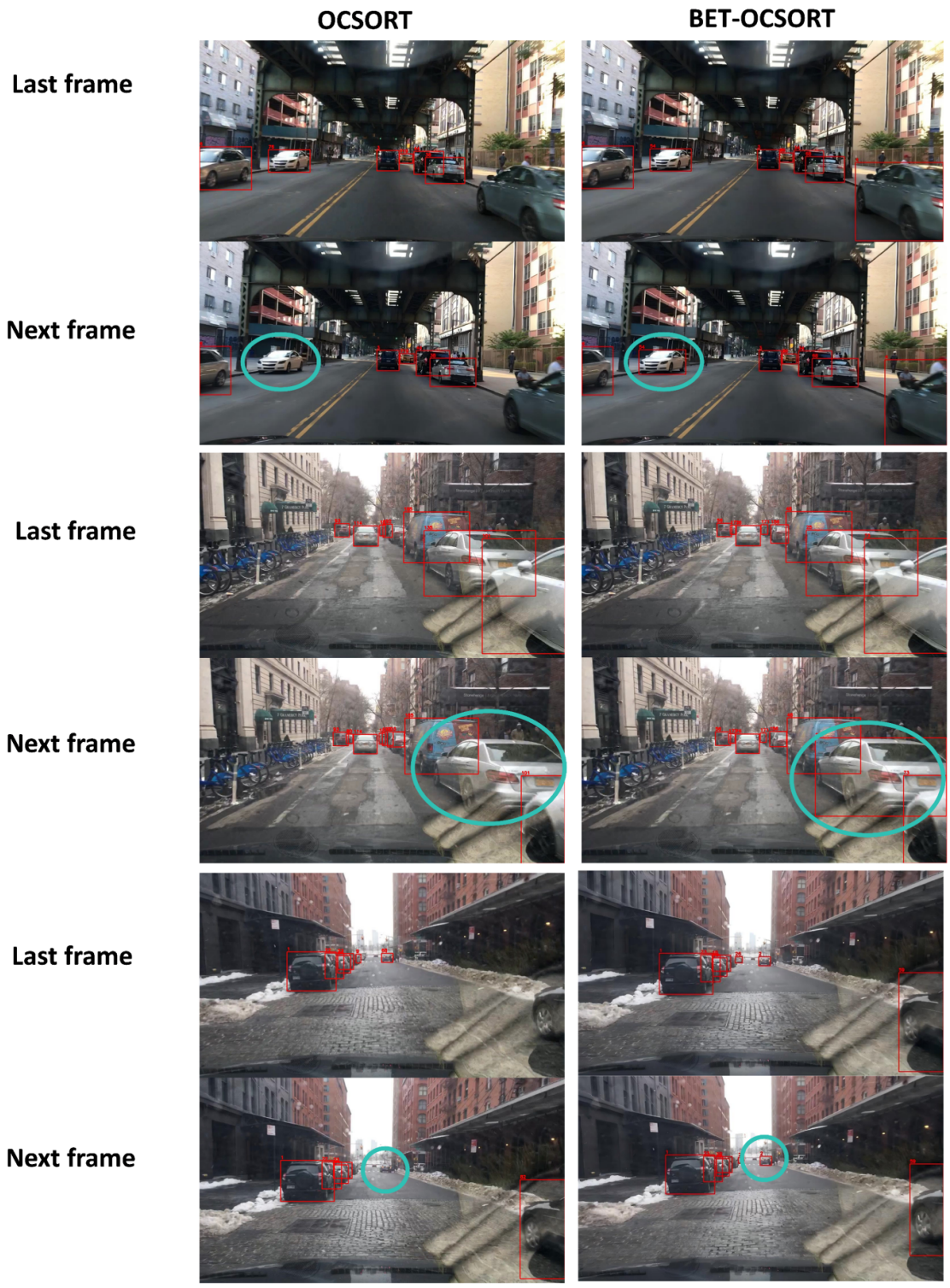


Figure 13. Visualization of tracking results on the BDD100K dataset.

make the association process more focused on reliability and identity stability. As a result, the proposed strategy may be relatively conservative for some marginal or low-confidence targets, leading to a slightly lower IDF1 score. However, it can effectively reduce erroneous associations and identity switches, thereby improving the overall stability of multi-object tracking.

Table 11. Comparative results on the KITTI dataset.

Model	MOTA	HOTA	IDF1	IDSW
DeepFusion-MOT(RCC + PC) [31]	40.20%	42.60%	44.90%	1203
DFR-FastMOT(RCC + PC) [31]	91.00%	81.90%	90.10%	215
EagerMOT(RCC + P-GNN) [31]	87.30%	78.00%	89.30%	91
EagerMOT(YOLOV3 + PC) [31]	41.60%	36.50%	48.10%	792
DeepOCSORT	90.27%	80.01%	86.00%	145
StrongSORT	93.16%	80.52%	85.04%	352
SORT	87.97%	66.75%	72.00%	298
HybridSORT	70.91%	53.85%	50.47%	641
ByteTrack	91.75%	81.00%	93.54%	177
Ours	92.28%	83.69%	91.73%	99

5. Conclusion

This paper proposes a two-stage vehicle detection and tracking algorithm for autonomous driving scenarios. An improved object detection network, LDA-YOLO26, is developed to achieve higher detection accuracy with lower FLOPs, thereby striking a favorable balance between detection performance and computational efficiency. Experiments conducted on the KITTI and BDD100K datasets verify the positive contributions of the proposed modules and demonstrate the superiority of the improved network over existing mainstream detection networks on the KITTI dataset. Subsequently, this network is employed as the feature extraction network to provide feature input for the tracking model. The proposed BET-OCSORT tracking framework introduces ECC-based camera motion compensation to correct viewpoint variations caused by vehicle vibrations, improves the Kalman filter by dynamically adjusting the noise level to achieve more accurate prediction, and finally adopts TAI to strictly control the initialization of new trajectories, thereby improving the initialization quality and subsequent association stability in online multi-object tracking. Evaluation and visualization experiments on the KITTI and BDD100K datasets show that the improved tracking model achieves high tracking accuracy within the two-stage tracking framework.

Although the proposed method has achieved significant results, there is still room for improvement. In extremely complex scenarios, such as heavy snow and haze, the detection model can perform preliminary recognition, but its detection precision and accuracy still need to be further improved; this could be addressed through task-specific training in future work. In the tracking stage, although the current model has achieved a relatively high level of performance, its processing speed still needs to be improved, and its real-time capability under resource-constrained conditions requires further optimization.

Vehicle detection and tracking technology provides important technical support for path planning in autonomous driving. Similar tasks can also be extended to robotics as well as obstacle recognition and path planning for unmanned aerial vehicles. Overall, this research area is of great significance and holds considerable potential.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Conflict of interest

The authors declare no conflicts of interest.

Ethics Statement

This study does not involve human or animal subjects.

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